

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and owning distributed infrastructure at scale. Built systems processing billions of events daily at YouTube and Google, including foundational contributions to **Bigtable**, **Spanner**, and internal ML platforms. Led teams across infrastructure, data, and search domains while partnering with cross-functional teams to deliver platform reliability. Focuses on distributed systems architecture, on-call driven development, and end-to-end ownership of tier-1 services. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Distributed Systems: Bigtable, Spanner, MapReduce, Kubernetes, Borg, internal cluster management

Storage and Data: BigQuery, Pub/Sub, GFS, MySQL, Vitess, Kafka

Platform and Infra: GCP, Terraform, Linux, on-call, SLO/SLAs, capacity planning

ML and Analytics: TensorFlow, internal ML infrastructure, feature pipelines, data warehousing

Languages and Tooling: Go, C++, Python, Java, Bazel, TypeScript, JavaScript, HTML, CSS

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Own platform reliability for video serving infrastructure handling billions of daily requests, driving on-call driven development across tier-1 services.
- Designed multi-region container migration strategy reducing infrastructure costs while maintaining strict SLOs for 250M+ active users.
- Led migration of batch workloads from legacy processing to **Kubernetes** infrastructure, improving resource utilization by 40%.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built event processing pipeline handling 10M+ events per second using **Kafka** and **Pub/Sub** for real-time inventory management.
- Architected compute optimization pipeline migrating core services to **Kubernetes**, reducing infrastructure spend by \$5M annually.
- Owned end-to-end reliability for marketplace platform supporting \$1B+ in annual transactions with sub-second latency requirements.
- Established on-call rotation and incident response framework for 15-engineer infrastructure team.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of 30 engineers across YouTube search infrastructure, hiring 24 engineers and driving platform reliability from 99.9% to 99.99% uptime.
- Led migration of YouTube search ranking data from **MapReduce** to **Bigtable**, reducing query latency by 60% for 2B+ users.
- Designed and shipped **Spanner** schema migration tooling used by 200+ services, eliminating planned downtime for global database operations.
- Defined and enforced SLOs for serving infrastructure across 12 product teams, centralizing on-call ownership.
- Owned YouTube recommendation feature pipeline processing 500B+ daily examples for **TensorFlow** model training.

Software Engineer | Google | 02/2011 - 02/2015

- Built automated capacity planning system for search infrastructure, predicting hardware needs 90 days ahead with 95% accuracy.
- Contributed to **Bigtable** internal storage layer, implementing compaction algorithms handling 1PB+ daily writes across 50K+ machines.
- Developed kernel-level profiling tools for **Linux** distributed systems, reducing p99 latency by 35% in production.
- Owned MapReduce infrastructure migration for AdWords reporting, consolidating 200+ legacy pipelines into 12 scalable jobs.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed distributed data processing pipeline for Bing using **MapReduce**, indexing 50B+ web pages with sub-hour freshness.
- Built **MySQL** sharding framework reducing database query latency by 70% for search serving at scale.
- Developed internal workflow orchestration tool adopted by 8 product teams, replacing manual deployment processes.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Implemented distributed sorting algorithm for internal **MapReduce** framework, processing 10TB datasets in under 2 hours.
- Optimized **Linux** kernel network stack for high-throughput data transfer between storage nodes.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007